🔭 1. The Copernican Revolution (Astronomy)

Key Figures & Models
Claudius Ptolemy (Geocentric Model): Proposed that the Earth is stationary at the center of the universe, and all heavenly bodies revolve around it. It used deferents (large circular orbits) and *epicycles* (smaller circles centered on the deferent) to explain *retrograde motion*.

Nicolaus Copernicus (Heliocentric Model): Argued that the Sun is at the center and planets revolve around it.

Major Work: De revolutionibus orbium coelestium (*On the Revolutions of the Heavenly Spheres*), published in 1543 just before his death.

Core Postulates:

1. The Earth is not the center of the universe, only the center of gravity and the lunar orbit.
2. The Sun is stationary near the center; planets revolve around the Sun in circular orbits (Mercury, Venus, Earth, Mars, Jupiter, Saturn).
3. The Earth has more than one motion: daily axial rotation and annual revolution around the Sun.

Johannes Kepler: Discovered that planetary orbits are actually elliptical, not perfect circles.
Galileo Galilei (1610): Used the telescope to provide observational proof against geocentrism: Discovered the four largest moons orbiting Jupiter (showing not everything revolves around Earth). Observed the phases of Venus (proving it orbits the Sun).

🧬 2. The Darwinian Revolution (Evolution)

Core Concepts & Influences
Definition of Evolution: Gradual change in organisms over generations from ancestral forms through the mechanism of natural selection.

Major Works:

- **On the Origin of Species (1859):** Introduced natural selection and descent with modification.
- **The Descent of Man (1871):** Applied evolutionary theory to human origins.
- **The Expression of the Emotions in Man and Animals (1872):** Explored shared emotional behaviors across species.
- **Thomas Malthus:** Influential political economist whose essay argued that human population grows geometrically (exponentially), whereas food production grows arithmetically. This insight led Darwin to realize there is a constant struggle for existence.
- **Herbert Spencer:** Coined the phrase "survival of the fittest" after reading Darwin's work.Evidence for Evolution
- **Homologous Structures:** Body parts sharing a common evolutionary origin and anatomical plan, even if used for different functions (e.g., human arm, cat leg, whale flipper, bat wing).
- **Analogous Structures:** Structures with similar functions but dissimilar evolutionary origins and anatomy (e.g., wings of birds, bats, and insects).

- **Cytochrome c & DNA:** Highly conserved proteins and genetic sequences across related species demonstrate biochemical evidence of common ancestry.

🧠 **3. The Freudian Revolution (Psychoanalysis)**

Foundations
Background: Sigmund Freud originally trained in neurology before developing psychoanalysis.

- **Josef Breuer & Studies in Hysteria (1895):** Included the famous case of Anna O., introducing the "talking cure" and the study of physical symptoms without organic causes (*hysteria*).
- **Goal of Psychoanalysis**: To bring repressed unconscious desires, conflicts, and memories into conscious awareness.
- **Erik Erikson:** Unlike Freud (who focused on childhood psychosexual stages), Erikson emphasized psychosocial development across the entire lifespan.

Structural Model of the Mind
Id: Operates on the pleasure principle; instinctual drives (eros/libido and thanatos).
Ego: Operates on the reality principle; mediates between the demands of the Id, Superego, and reality.
Superego: Operates on the morality principle; internalizes societal rules and ideals.

Freud's Psychosexual Stages

1. **Oral Stage** (0–18/21 months): Pleasure centered on the mouth (sucking, biting).
2. **Anal Stage** (18 months–3 years): Bowel and bladder elimination.
3. **Phallic Stage** (3–6 years): Genital exploration; discovery of anatomical differences between sexes; Oedipus/Electra complexes.
4. **Latency Stage** (6 years–puberty): Dormant sexual feelings; focus on school and socialization.
5. **Genital Stage** (puberty onward): Maturation of sexual interests.

👥 **4. Influences & Key Figures (Copernicus, Darwin, & Freud)**

1. Aristarchus of Samos

Role/Contribution: *An ancient Greek astronomer* who presented the earliest known heliocentric model of the universe.

Key Finding: Examined the relative sizes and distances of the Earth, Sun, and Moon, calculating that the Sun is roughly 20 times farther away than the Moon and around 20 times larger (and thus too large to orbit a tiny Earth).

Worksheet Match: Aristarchus

2. Aristotle
Role/Contribution: *Ancient Greek philosopher* whose geocentric worldview dominated Western thought for nearly two millennia.

Key Concept: Taught that the heavens were unchanging and perfect, asserting that all celestial bodies are flawless, perfect spheres with absolutely smooth surfaces moving in uniform circular paths.

Worksheet Match: Aristotle

3. Charles Lyell
Role/Contribution: Scottish geologist and author of *Principles of Geology*, which deeply influenced Charles Darwin aboard the HMS Beagle.

Key Concept: Championed uniformitarianism—the idea that the Earth is extremely old and that geological features changed gradually through time via slow, continuous natural processes (erosion, sedimentation, volcanic activity) rather than sudden biblical catastrophes.

Worksheet Match: Charles Lyell

4. Sir Isaac Newton
Role/Contribution: English physicist and mathematician whose work united terrestrial and celestial physics.

Key Concept: Formulated the Universal Law of Gravitation and laws of motion, showing that the exact same mathematical principles govern the trajectory of an apple falling from a tree and the orbit of the Moon around Earth.

Worksheet Match: Isaac Newton

5. Dr. Erasmus Darwin
Role/Contribution: Charles Darwin's grandfather; a prominent English physician, poet, philosopher, and inventor.

Key Concept: One of the earliest British thinkers to propose that all warm-blooded animals arose from a single living filament, writing early evolutionary ideas in works like Zoonomia that served as an early intellectual influence for Charles Darwin.

Worksheet Match: Dr. Erasmus Darwin

🛡️ **5. Ego Defense Mechanisms (Quick Matcher)**

These unconscious mechanisms protect the ego from anxiety caused by inner conflicts:

Defense Mechanism	Identity	Worksheet Example
Projection	Attributing one's own unacceptable impulses or traits to others.	Criticizing others for gossiping when you gossip; believing your boss dislikes you when you actually dislike your boss.
Regression	Reverting to an earlier, less mature developmental stage during stress.	An adult returning to live with parents and acting like a dependent child after marital failure.
Reaction Formation	Expressing the exact opposite of what you genuinely feel.	Going out of your way to show love and warmth to someone you despise.
Undoing	Engaging in a behavior to magically cancel out or atone for a previous wrongdoing.	A corrupt businessman donating heavily to charity to compensate for exploiting

		others.
Repression	Involuntarily pushing painful, traumatic memories out of conscious awareness.	Completely forgetting details of a severe childhood car accident.
Denial	Refusing to accept an obvious reality or painful fact.	Insisting a medical diagnosis of chronic illness is incorrect.
Rationalization	Creating logical-sounding excuses to justify failure or unacceptable outcomes.	Saying "I didn't want that job anyway" after being rejected.
Displacement	Redirecting anger or impulses toward a safer, less threatening substitute target	A batter striking out and taking anger out on the water cooler.
Sublimation	Channeling unacceptable impulses or emotional pain into productive, creative activities.	Writing songs or making art to process heartbreak or anger.